

Neural Networks for NLP - Sheet 4

06.01.2025

Deadline: 20.01.2025 - 13:15, on Olat as **.pdf/.html/.ipynb + .txt** and/or after the lecture

Task 1: Reinforcement Learning on Human Feedback

16 Points

The goal of this task is to develop a language model that generates happy tweets. For this task you should use the `miniGPT` model (provided), which you should pre-train and fine tune using RLHF with PPO. To this end, you should follow the instructions in the `happyGPT.ipynb` notebook to

1. Pre-train the `miniGPT` model. (2 points)
2. Implement the reward model to be used in the RL training process. (4 points)
3. Implement the RL fine tuning process using PPO. (4 points)
4. Run the code to fine tune the `miniGPT` model and briefly comment on the quality of the responses. Do they look 'happy'? How do they compare to the responses after pre-training in terms of sentiment? (4 points)
5. Given the training process of the `InstructGPT` model, what step would you add in the above training process of the `miniGPT` model to improve it? (2 points)

Task 2: Reinforcement Learning for Language Understanding

8 Points

Our task is to develop a reinforcement learning (RL) agent that uses deep Q-learning to play text-based games, such as Zork1. The primary aim of the agent is to consistently outperform human players in achieving higher scores. During gameplay, the agent will receive descriptions of the current state, a list of possible actions, and the current score from the game designer after each move. According to the received feedback/reward, the RL agent will determine the most optimal action from the list of possibilities.

Suppose the game begins with the player starting in an empty room. The RL agent receives a description from the game environment: "You find yourself in an empty room and notice a door on the east side." The possible actions the agent can take are "east", "north", "west", and "south". If the agent selects the action "move east", it will transition to a new position and receive updated description and actions. If it chooses any other action, the agent will remain in its starting position. The agent will receive a score after each move. An example is shown below.

Please answer the questions and justify your answer:

1. Is Q-learning considered an on-policy or off-policy method, and could you explain it? (1 point)
2. Why is the trade-off between Exploration and Exploitation hard? (1 point)
3. Define the Markov Decision Process (MDP) tuple for the above problem. List at least one possible transition example with a random number ($\gamma = 0.9$). (1 point)

Description (start point): You find yourself in an empty room and notice a door on the east side.

Valid actions space: east(e), north(n), west(w), south(s)

Prediction: east

Score: 0

Description: You step into a beautiful garden. You notice an old book lying on a nearby bench.

Valid actions space: take the book, east, west, north

Prediction: take the book.

Score: 1

4. Provide an overview of one possible deep Q-learning neural network architecture, including a brief explanation of any relevant embedding technique. Which function can be used to approximate the $Q(s, a)$ value? (3 points)
5. Assuming the predicted Q-values by the policy for different actions at the next state: $Q(s_2, \text{take the book}) = 0.2$, $Q(s_2, \text{east}) = 0.4$, $Q(s_2, \text{west}) = 0.3$, $Q(s_2, \text{north}) = 0.1$. If the agent goes 'east,' it will receive a reward of 1, while other actions yield a reward of 0. Provide the mathematical formulation of the Q-learning update rule and compute the optimal Q-value. (2 points)

General rules

1. You need to complete and run the entire code in the notebook given to you as part of this assignment. Submitting the completed notebook containing your results is **mandatory**. For theoretical answers, you may add them in the notebook (in text cells) or you can also submit your answers to such questions in a separate .pdf file.
2. Any kind of plagiarism and cheating (from your classmates as well as from the internet) is **strictly forbidden** and will lead to your disqualification from the course.
3. You are asked to submit your answers as .pdf/.html/.ipynb and additionally your output as a .txt file.